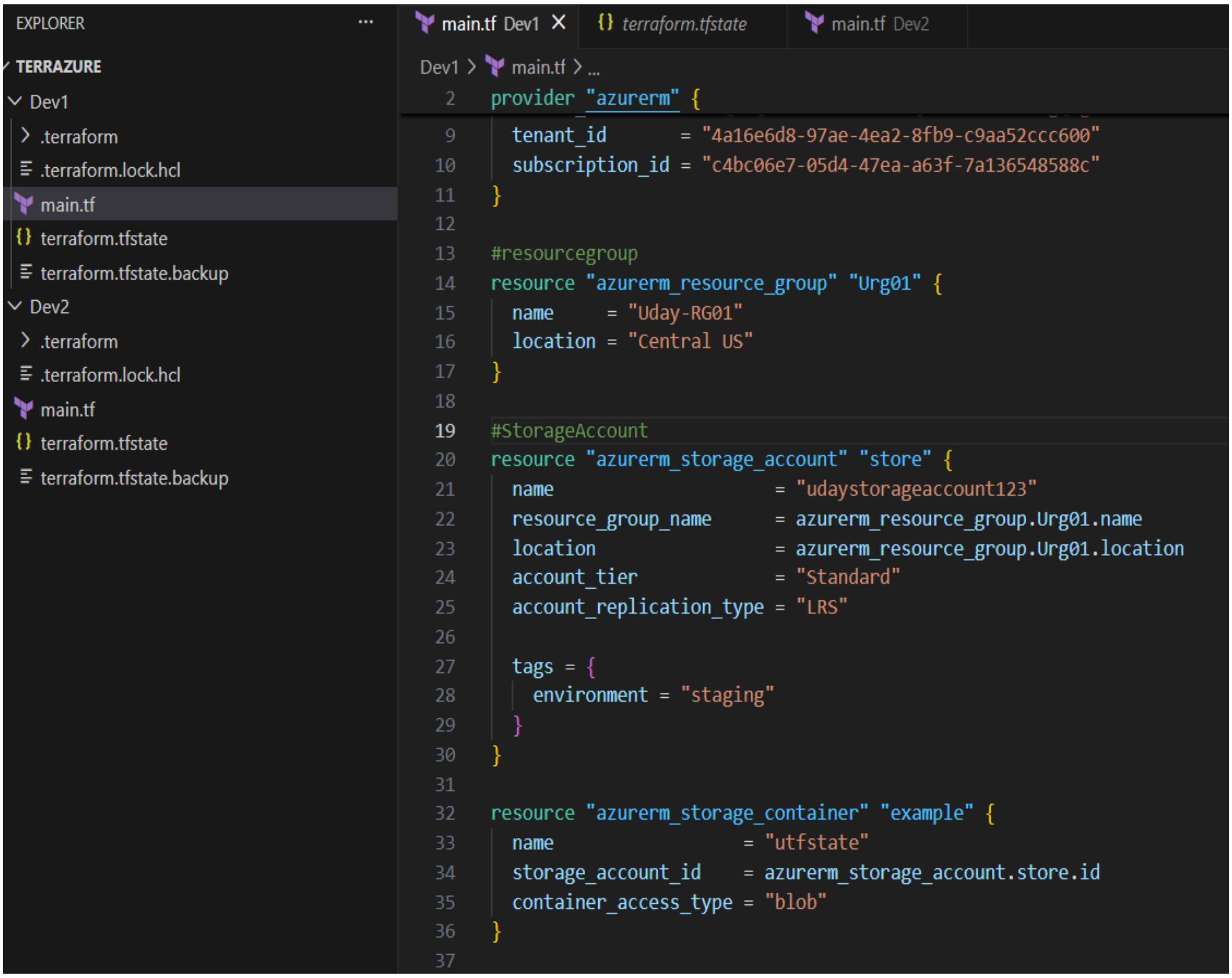
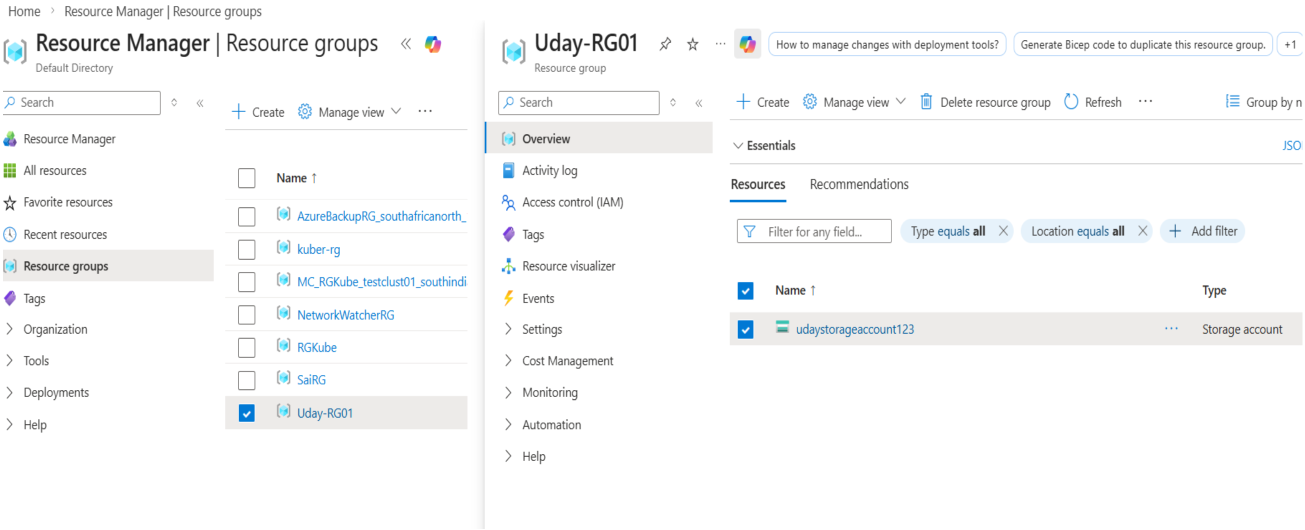
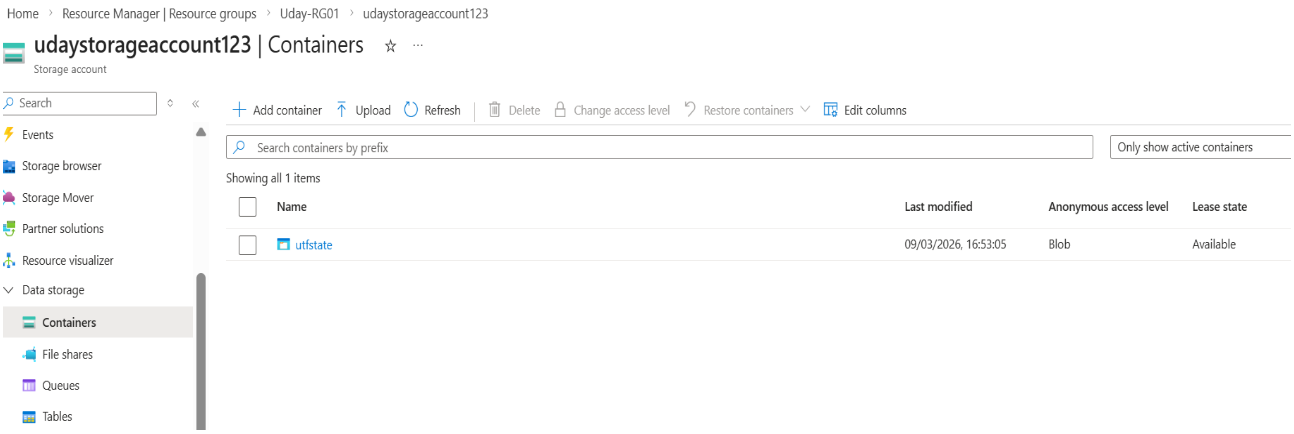
3. Terraform Backend Block

- ✓ **Statefile should be in centralized location (Storage Account).**
- ✓ **Every user can use that statefile and they can create resources using that.**
- ✓ **We can use same code and create resources in different environments, but for that we should create different workspaces and use different statefiles.**
- **Below I have created storage account and stored statefile inside container**



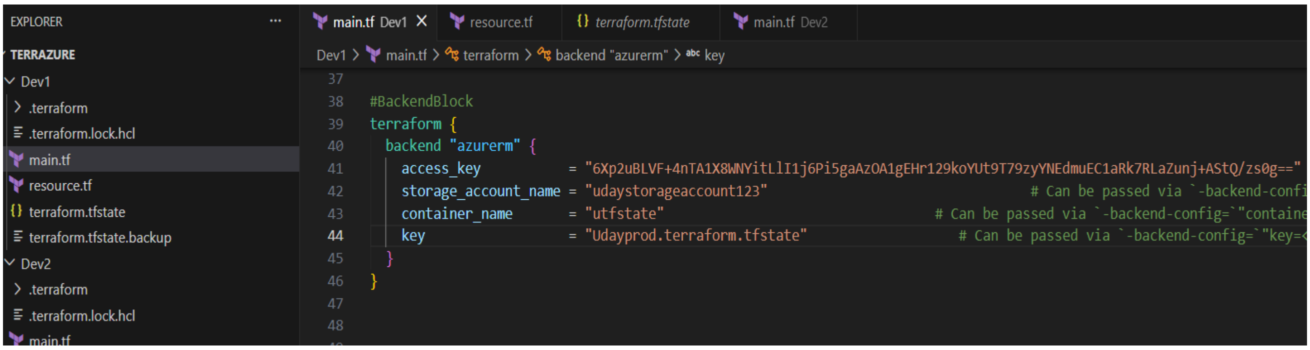

```
provider "azurerm" {
  tenant_id       = "4a16e6d8-97ae-4ea2-8fb9-c9aa52ccc600"
  subscription_id = "c4bc06e7-05d4-47ea-a63f-7a136548588c"
}

#resourcegroup
resource "azurerm_resource_group" "Urg01" {
  name     = "Uday-RG01"
  location = "Central US"
}

#StorageAccount
resource "azurerm_storage_account" "store" {
  name                     = "udaystorageaccount123"
  resource_group_name      = azurerm_resource_group.Urg01.name
  location                 = azurerm_resource_group.Urg01.location
  account_tier             = "Standard"
  account_replication_type = "LRS"

  tags = {
    environment = "staging"
  }
}

resource "azurerm_storage_container" "example" {
  name                  = "utfstate"
  storage_account_id    = azurerm_storage_account.store.id
  container_access_type = "blob"
}
```

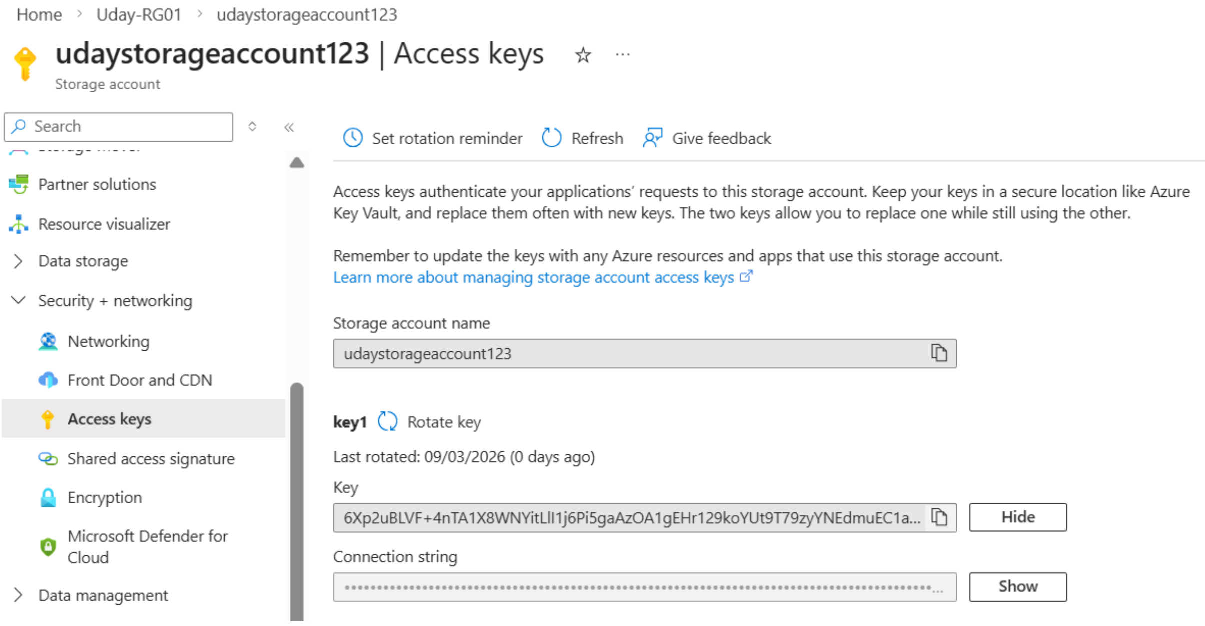




Next configure BackendBlock from local to storage.

- **For that we need below 4.**

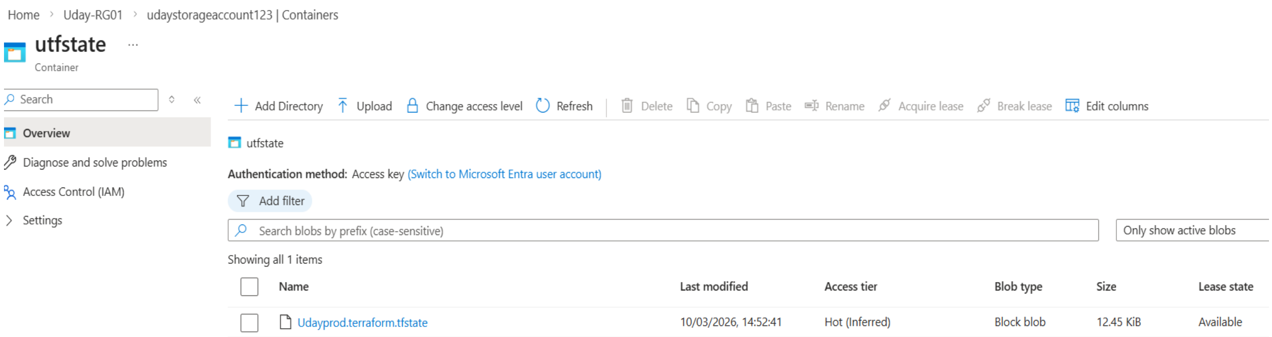




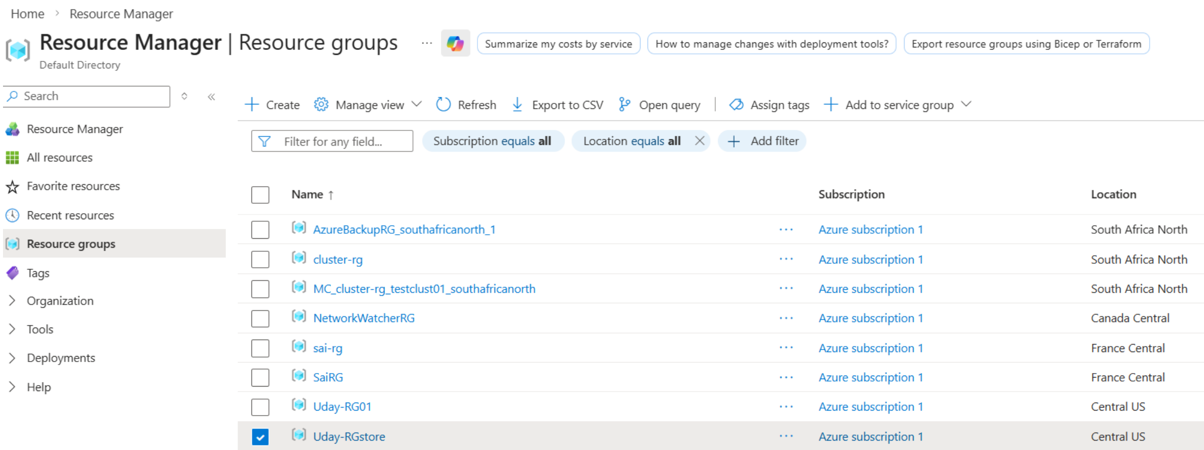
✓ **Let's run terraform init, then it will ask for conformation to copy existing state to new backend.**

```
PS C:\TerrAzure> cd C:\TerrAzure\Dev1\
PS C:\TerrAzure\Dev1> terraform init
Initializing the backend...
Acquiring state lock. This may take a few moments...
Do you want to copy existing state to the new backend?
  Pre-existing state was found while migrating the previous "local" backend to the
  newly configured "azurerm" backend. No existing state was found in the newly
  configured "azurerm" backend. Do you want to copy this state to the new "azurerm"
  backend? Enter "yes" to copy and "no" to start with an empty state.

  Enter a value:
```

✓ **After YES, above key created below in statefile.**



✓ **Now let's create new RG,**



✓ **Whatever resources created it will store in storage account statefile and disapper in local statefile.**

- **Below resources are available in statefile**

```
PS C:\TerrAzure\Dev1> terraform state list
azurerm_resource_group.Urg01
azurerm_resource_group.Urgstore
azurerm_storage_account.store
azurerm_storage_container.example
```

- **Above I want to remove particular resource from statefile but not from portal.**
- **Don't remove code from mainfile, if we do that the resources will delete from portal and from statefile.**
- **I don't want to delete any code, I just want to remove resources from statefile and delete other resource.**

✓

```
PS C:\TerrAzure\Dev1> terraform state
Usage: terraform [global options] state <subcommand> [options] [args]

  This command has subcommands for advanced state management.

  These subcommands can be used to slice and dice the Terraform state.
  This is sometimes necessary in advanced cases. For your safety, all
  state management commands that modify the state create a timestamped
  backup of the state prior to making modifications.

  The structure and output of the commands is specifically tailored to work
  well with the common Unix utilities such as grep, awk, etc. We recommend
  using those tools to perform more advanced state tasks.

Subcommands:
    identities          List the identities of resources in the state
    list                List resources in the state
    mv                  Move an item in the state
    pull                Pull current state and output to stdout
    push                Update remote state from a local state file
    replace-provider    Replace provider in the state
    rm                  Remove instances from the state
    show                Show a resource in the state
PS C:\TerrAzure\Dev1>
```

```
PS C:\TerrAzure\Dev1> terraform state list
azurerm_resource_group.Urg01
azurerm_resource_group.Urgstore
azurerm_storage_account.store
azurerm_storage_container.example
```

✓ **Let’s remove container & storage,Rg-store (to remove dependency for storage account) account from statefile.**

```
PS C:\TerrAzure\Dev1> terraform state rm azurerm_storage_container.example
Acquiring state lock. This may take a few moments...
Removed azurerm_storage_container.example
Successfully removed 1 resource instance(s).
Releasing state lock. This may take a few moments...
```

```
PS C:\TerrAzure\Dev1> terraform state rm azurerm_storage_account.store
Acquiring state lock. This may take a few moments...
Removed azurerm_storage_account.store
Successfully removed 1 resource instance(s).
Releasing state lock. This may take a few moments...
```

```
PS C:\TerrAzure\Dev1> terraform state rm azurerm_resource_group.Urgstore
Acquiring state lock. This may take a few moments...
Removed azurerm_resource_group.Urgstore
Successfully removed 1 resource instance(s).
Releasing state lock. This may take a few moments...
```

✓ **After removing, we have only 1 below.**

```
PS C:\TerrAzure\Dev1> terraform state list
azurerm_resource_group.Urg01
```

- **Now if we remove above 3 resources from code and when we run terraform apply, it will show infra-matchs below result.**

```
PS C:\Users\kucha\Desktop\Azureterra\Dev1> terraform apply
Acquiring state lock. This may take a few moments...
azurerm_resource_group.rg01: Refreshing state... [id=/subscriptions/0cf71a42-7c78-413c-aa9a-e3adaf9c9972/resourceGroups
standard-rg]
azurerm_virtual_network.vnet01: Refreshing state... [id=/subscriptions/0cf71a42-7c78-413c-aa9a-e3adaf9c9972/resourceGro
ps/standard-rg/providers/Microsoft.Network/virtualNetworks/vnet0111]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are
needed.
```

❖ ACQUIRING STATE LOCK

```
PS C:\Users\kucha\Desktop\Azureterra\Dev2> terraform destroy
Acquiring state lock. This may take a few moments...
azurerm_resource_group.rg01: Refreshing state... [id=/subscriptions/0cf71a42-7c78-413c-aa9a-e3adaf9c9972/resourceGroups
standard-rg]
```

- **When 2dev are working, this acquiring state lock will be applied one after another, so that only one dev will work at a time.**